Solving the Invigilation Scheduling Problem in Examination Timetabling System Using Tabu Search Technique

By

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DECLARATION

The project report submitted herewith is a result of my own efforts in totality and in every aspects of the works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion of any sorts constitutes a breach of Tunku Abdul Rahman University College rules and regulations and would be subjected to disciplinary actions.

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APPROVAL FOR SUBMISSION

I certify that this project report entitled **"SOLVING THE INVIGILATION SCHEDULING PROBLEM IN EXAMINATION TIMETABLING SYSTEM USING TABU SEARCH TECHNIQUE"** was prepared by **LIONG JEE YUEN** and has met the required standard for submission in partial fulfilment of the requirements for the award of Bachelor of Science (Hons.) Management Mathematics with Computing at Tunku Abdul Rahman University College.

Approved by,

Signature: ____________________ **Signature:** ____________________

Supervisor: Dr. Chin Wan Yoke **Moderator:** Dr. Tan Yan Bin

Date: ____________________ **Date:** ____________________

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ABSTRACT

In this study, the invigilation duty scheduling problem in the examination timetabling system of Tunku Abdul Rahman University College (TAR UC) will be solved by using the Tabu Search technique. The purpose of conducting the project is to investigate the linear relationship between the size of Tabu List and the performance of Tabu Search and thus determining the most appropriate choice of the size of Tabu List during the implementation of the algorithm. Besides that, the project also aims to investigate the performance of Tabu Search in solving the invigilation scheduling problem. Furthermore, a comparison between the performance of Tabu Search and the current algorithm that was implemented in the examination timetabling system of TAR UC will be carried out in order to determine the suitability of both of the algorithms in solving this problem. In order to achieve these objectives, the implementation of Tabu Search with different size of Tabu List in the invigilation duty scheduling problem will be done and followed by a series of tests to be conducted. At the same time, tests will also be conducted on the invigilation module in the examination timetabling system of TAR UC and the results will be compared to the results from Tabu Search algorithm. Based on the results, the performance of Tabu Search is not linearly related to the size of Tabu List and therefore, the implementation of Tabu Search with Tabu List's size of 20 has been chosen for further analysis. From the study, Tabu Search can be considered as an efficient algorithm in solving the invigilation duty scheduling problem but lack of consistency in maintaining the high efficiency all the time. However, as compared to the current algorithm, Tabu Search is able to outperform the current algorithm in a great extent in terms of efficiency as well as consistency in solving the problem. Therefore, this study has concluded that Tabu Search has a great potential in solving the invigilation duty scheduling problem as it has a high efficiency in solving the problem despite of the low consistency in maintaining the performance.

Keywords: invigilation scheduling, Tabu Search, Tabu List, linear relationship, performance, suitability, efficiency, consistency

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CHAPTER 1 Introduction

1.1 Introduction

Examination timetabling is a process that allocate all of the examinations of a particular institution into available rooms and timeslots that have been provided without producing any conflict between the examinations. Besides that, this process also includes the design of floor plan as well as allocation of invigilators for each of the examination that will be conducted. It could be a very complicated and tedious task especially for the higher educational institutions as they will have to deal with a large number of examinations with limited venues and time. Therefore, there are several research have been done on the development of automated examination timetabling system since a few decades ago. An examination timetabling system is a great tool which help to automate the whole process of examination scheduling for an institution.

In this study, we will focus on the invigilation duty scheduling problem which is a problem of allocating the staff of a particular education institution to all of the invigilation duty according to different constraints that are stated. Throughout the project, this problem will be solved by using the Tabu Search algorithm which is a metaheuristic that is flexible in solving various kinds of optimization problem. With this study, the efficiency of Tabu Search in solving this particular problem will be evaluated so that the suitability of this approach in the invigilation duty scheduling problem can also be determined. Furthermore, the system created with Tabu Search approach and the current system will be evaluated by using the data provided by Tunku Abdul Rahman University College in order to compare the performance of both of the algorithm and thus determine the better choice in solving the scheduling problem.

1.2 Problem Statement

The examination timetabling system of Tunku abdul Rahman University College (TAR UC) is currently facing two main problems, which are the low efficiency in generating the invigilation duty schedule as well as the performance inconsistency of the system.

The performance of the current examination timetabling system of TAR UC was unsatisfying due to the high computational time in scheduling the invigilation duty. The current system require a large amount of time to produce a feasible invigilation duty schedule for the college. This problem will also lead to power consuming as the computer system of the college have to remain power on for a long period of time. Therefore, this problem should not be overcame and the performance of the system should be improved in terms of computational time.

Besides that, the performance of the examination timetabling system of TAR UC is not consistent in terms of the quality of schedule as well as the time take to generate a feasible schedule. Both of these aspects are important criteria to determine the quality of an examination timetabling system. However, the current system was unable to guarantee a high quality schedule for every scheduling process that was carried out. Besides that, the computational time that required to produce a feasible result is hard to be predicted as the time taken by the system was inconsistent

1.3 Objectives

The main purpose of carrying out this project is to apply the Tabu Search Technique in solving the invigilation duty scheduling problem of the current examination timetabling system which is still under development. This project aims to determine the efficiency of Tabu Search in solving the particular optimization problem as mentioned above. The scheduling algorithm of the system will be modify in such a way that all of the hard constraints, which have been stated by the Department of Examinations and Credit Accumulation (DECA), are being satisfied. Besides that, the soft constraints that have been mentioned will also be included into the consideration of this project in order to produce a better solution and thus improve the satisfaction of the relevant party.

Besides that, this project wish to optimise the performance of the invigilator module of the current examination timetabling system. The study aims to minimized the computational time of the system in generating a invigilation duty schedule without overlooking any of the constraints that have been stated. Moreover, the consistency of the system will also be one of the main concern of the project so that the system will be able to produce an acceptably good schedule with an appropriate time length consistently. Therefore, the project will also aim to compare the performance and suitability of the Tabu Search algorithm and current algorithm in order to determine the suitable algorithm to be applied in this particular problem.

1.4 Significance of the Study

Through the study, the main advantage that can be achieved is to be able to determine the efficiency and suitability of Tabu Search algorithm in solving the invigilation duty scheduling problem. Nowadays, there are various kinds of metaheuristic algorithms that have been discovered in order to solve different types of optimization problems. However, different approach gives different effect in a particular optimization problem in terms of performance and efficiency. Therefore, in order to solve an optimization problem more efficienctly and effectively, the most suitable metaheuristic have to be chosen instead of simply applying any of the algorithm. Therefore, by applying Tabu Search technique to the problem as mentioned previously, its suitability in the problem can be determined. Besides, this could also act as a reference for the others who are desired to solve any optimization problem that are similar with the invigilation duty scheduling problem as this could help them to understand the application of Tabu Search in this kind of problem clearer.

Apart from that, the performance of the current system might be improved in terms of quality and computational time through the application of different approach in the scheduling of the invigilation duty. In the way of finding a better solution in solving the invigilation duty scheduling problem of the examination timetabling system in TAR UC, the discovery of the most suitable metaheuristic in the problem is a critical issue. Therefore, the application of Tabu Search approach might give a chance in improving the performance of the current system with a great extent due to its flexibility in solving various types of optimization problems. Consequently, the improvement in the performance of the current system could help the college to save up cost and time in generating a complete examination timetable as well as invigilation duty schedule.

CHAPTER 2 Literature Review

2.1 Recent Papers/ Discoveries :

2.1.1 Comparison of Metaheuristics in Various Problems

Throughout the years, there were several comparisons that have been made on the performance of different types of metaheuristic in order to provide a guideline for the users in selecting the most suitable approach to solve a specific combinatorial optimization problem. In 1999, Chu and Fang have carried out an investigation on the application of Tabu Search and Genetic Algorithm on timetabling problem. The performance of both algorithms was compared between each other and the result showed that Tabu Search could perform better than Genetic Algorithm as it was able to produce a better solution within a shorter period of computational time compared to Genetic Algorithm.

In the year of 2002, an unbiased comparison of the performance in solving university course timetabling problem have been carried out among five metaheuristics which includes the Evolutionary algorithm, Ant Colony Optimization, Iterated Local Search, Simulated Annealing and Tabu Search. In order to ensure the fairness of the comparison, the algorithms used common solution representation throughout the experiment and all of the test will be conducted under the same condition where the algorithms will be run in the same compiler as well as hardware. The result of the experiment showed that only Iterated Local Search and

Tabu Search were having chance to generated feasible solutions to the problem with large instances. (Rossi-Doria *et al.*, 2002)

Few years later, Azimi (2004) had also solved the examination timetabling problems by using the Tabu Search, Genetic Algorithm, Simulated Annealing and Ant Colony System in their basic framework. The results generated with ten datasets by four of the algorithms were then used to compare among each other. Based on the comparison, Ant Colony System works the best in this problem and followed by Tabu Search. This is because Ant Colony System manage to generate a better initial solution as compared to the others which generate the initial solution in a random manner. However, the result also showed that Tabu Search was able to reduce the cost in generating solution with the greatest extent among all of the algorithms that involved in this comparison.

Besides that, Arostegui Jr *et al.* had compared the performance of Tabu Search, Genetic Algorithm and Simulated Annealing on different types of facilities location problems under time-limited, solution-limited and unrestricted condition in the year of 2006. As a result, Tabu Search showed the best performance in most of the cases while Genetic Algorithm and Simulated Annealing could only perform well in certain cases. Through the comparison, the authors claimed that Tabu Search has the higher chance to provide a better solution as compared to the Genetic Algorithm and Simulated Annealing, and therefore, it should be considered as the first choice in the facilities location problem due to its outstanding performance as well as the ease of its implementation.

In overall, Tabu Search is a flexible approach as it showed an acceptably good performance in a variety of combinatorial optimization problems. Therefore, it is undeniable that Tabu Search has a strong potential in solving the examination timetabling problem with a superior performance.

2.1.2 Previous application of Tabu Search on Educational Timetabling Problems

Since the year before 20^{th} century, Tabu Search technique has been widely used in solving the educational timetabling problem which includes examination timetabling and course timetabling for various academic institutions around the world. In the year of 1991, Hertz solved both type of educational timetabling problems with a proposed global approach which consists of two Tabu Search based heuristic methods called Tabu Search for the timetabling sub problem (TATI) and Tabu Search for the grouping sub problem (TAG). Consequently, this approach has been successfully applied to large scale timetabling problem by producing good solution to the problem.

A few years later, Costa (1994) had also made use of the Tabu Search algorithm in solving the course timetabling for secondary school and high school. This approach had produced a sufficiently good solution with only considerable amount of time. The author claimed that the presented algorithm can be easily modified to deal with different scheduling environment. However, this study could not show a comparison of the performance among Tabu Search and other existing methods due to the absence of a standard test problems that can be used to evaluate the value of each algorithm.

Later, White and Xie (2000) examined on the examination timetabling problem and thus implemented Tabu Search algorithm with a move-based longer term memory for the purpose of enhancing the solution of the problem. The result showed that this strategy is able to improve the effectiveness of the search besides generating a better solution compared to the solution generated by the approach of using only short term memory. In the same paper, the importance of determining the length of long term memory was also a concern of the author. A quantitative analysis method is a crucial process to be conducted to estimate the length of long term memory so that the diversification of search space could be prevented.

Besides that, Aladag *et al*. (2009) have investigated the effect of implementing different neighbourhood structures on Tabu Search. In this study, the authors were proposing two new neighbourhood structures based on the combination of two types of move which are called simple and swap. A comparison of the performance has been made between the newly proposed structures and the existing structures that only based on one type of move which is simple and swap. The experiment came to a conclusion that one of the proposed neighbourhood structures was able to provide a good solution to the timetabling problem.

Apart from that, Di Gaspero and Schaerf (2000) had made use of different features on the Tabu Search algorithm and thus investigated the influence of those features to the performance of Tabu Search. From the result, the best strategy is making use of shifting penalty mechanism, a variable size tabu list, a dynamic neighbourhood selection as well as a heuristic initial state. In this study, the authors also compared the performance of these Tabu Search based algorithms and the algorithms that was published in previous literature in solving the particular problem by using a variety of data set. As a result, the presented algorithms was considered to have a moderate performance as compared to the others as they could only outperform the other algorithms in some of the cases.

In conclusion, Tabu Search seems to be a powerful and effective approach that was commonly used to solve educational timetabling problem especially for examination timetabling. It could undergo with certain modification on some of its element besides adding on with some features in order to shorten the computation time and produce a better solution to a problem. Moreover, appropriate adjustment on the algorithm based on the problem faced is important as it could affect the effectiveness as well as efficiency of the searching process.

2.1.3 Previous application of Tabu Search on the problems in other fields

Besides the application on educational timetabling problem, Tabu Search has been also applied on various combinatorial optimization problems in other fields. In 1996, Nowicki and Smutnicki introduced a new algorithm based on Tabu Search algorithm which is called Tabu Search Algorithm with Back Jump Tracking (TSAB) for the aim of solving a permutation flow shop problem which is to minimize the maximum make span in a flow shop problem. With the reduced structure of the neighbourhood and the use of a special method to calculate the make span, the proposed algorithms showed a good performance by producing good solution with short computational time.

Besides that, Toth and Vigo (2003) also presented a good strategy in solving the vehicle-routing problem by introducing the Granular Tabu Search which make use of a restricted neighbourhood that eliminate myopic move that involve only elements that not likely belong to high quality feasible solution. The computational testing which has been conducted in this study showed that the implementation of granular neighbourhood in Tabu Search result in a better efficiency and shorter computational time.

Demeester *et al.* (2010) had contributed to the medical field by tackling the problem of patient admission scheduling with the use of Tabu Search technique. In this paper, the authors hybridized Tabu Search with the token-ring approach to solve the problem and the implementation was tested and compared with integer programming by using a random generated data set. In result, this study was able to improve the performance of the admission scheduling significantly.

Furthermore, Gallego *et al.* (2013) had developed a searching algorithm based on Tabu Search by adding on strategic oscillation which enables the search to visit the infeasible solutions. The proposed algorithm was used to solve a maximally diverse grouping problem which is to group a set of elements in which the diversity of the elements in the groups are being maximized. After conducting extensive computational testing, the authors claimed that this method gave advantage to the search by increasing its efficiency and effectiveness.

In short, the potential of Tabu Search technique in solving different kinds of combinatorial optimization problems can be clearly seen through the applications of this algorithm as mentioned above. It is a flexible approach that can be modified in order to suit into various type of cases and thus solving the specific problem with better efficiency. Therefore, a proper use of the strategies that has been proposed by the authors could be helpful in enhancing the performance of the invigilation module in the examination timetabling system of TAR UC.

CHAPTER 3 Methodology

3.1 Tabu Search

3.1.1 General Description of Tabu Search

In this project, the invigilation scheduling problem of the examination timetabling system will be solved by using the Tabu Search technique which is a meta-heuristic approach that is able to provide a sufficiently good solution to the problem in a fast and efficient way. It is basically consisted of several important elements such as the move, neighbourhood, initial solution, searching strategy, memory, aspiration function as well as stopping rules. In order to prevent the search from being trapped with the local optimal, a recency-based short term memory which is called tabu list was introduced for the use of storing the attribute of the moves that are forbidden. However, a feature which is called aspiration function will also be introduced so that a forbidden move is allowed to performed if the aspiration function evaluate it as a move that could improve the solution (Nowicki and Smutnicki, 1996).

3.1.2 Procedure of Tabu Search

According to Glover (1990), the procedures of Tabu Search are as below :

Step 1 : An initial solution will be randomly generated.

Step 2 : A candidate list of move that can be applied on the current solution will be generated where the move is not a tabu move or it is able to satisfy the aspiration criteria.

Step 3 : The best admissible move will be chosen.

Step 4 : The tabu condition and aspiration criteria will be updated.

Step 5 : The process will be terminated when it reached the stopping criteria, else, back to Step 2.

A general view of the Tabu Search algorithm is shown in Table 1 (Azimi, 2004) :

```
s:=initial solution in X;
nbiter:=0; (*current iteration*)
bestiter:=0;
(*iteration when the best solution has been found*)
bestsol:=s;
(*best solution*) T:=0;
Initialize the aspiration function A;
While (f(s) > f_) and (nbiter-bestiter<nbmax) do
nbiter:=nbiter+1;
Generate a set V _ of solutions si in N(s) which are either
not tabu or such that A(f(s) >= f(si);
Choose a solution s_ minimizing f over V _;
Update the aspiration function A and the tabu list T;
If f(s_) < F(bestsol) then
bestsol:=S_;
bestiter:=nbiter;
s:=s_;
End While
```

Figure 3.1 The Tabu Search Algorithm

3.1.3 Stopping Criteria

In Tabu Search, the stopping criteria can be the following (Hertz *et al.*, 1995):

1. The number of iterations exceeds the maximum number of iterations allowed.
2. The number of iterations since the last improvement of the current best solution exceeds a certain limit.
3. An optimum solution has been obtained.
4. The neighbourhood of the current solution was not able to be found.

3.2 Invigilation Duty Scheduling Problem

3.2.1 Description

Invigilation duty scheduling problem is a problem of allocating the staff from a particular educational institution to invigilate all of the examination that will be carried out during a specific period. The main focus of the problem is about the allocation of the resources which are the staff on the invigilation duties in such a way that all of the hard constraints that have been stated are satisfied.

For the basic version of invigilation duty scheduling, all of the staff will be equally allocated to all of the invigilation duties without overlapping of staff on the same session, which mean that avoiding from allocating a staff to more than one duty in the same session. Besides that, the number of duty of each staff should be approximately balanced against each others.

3.2.2 Assumptions

There are a few assumption to be made in this study :

1. There is at least one staff that is able to satisfy all of the hard constraint of a particular invigilation duty.
2. The invigilation duties can be assigned to all of the staff equally.
3. Some of the data that provided by TAR UC is dummy data.
4. All the staff are assumed to be working in the same branch.
5. Campbell paper is not included in the examination.

3.2.3 Constraints

Besides of the basic constraints as mentioned above, there are a lot of hard and soft constraint that have to be considered according to the particular institution itself. The additional constraints that have been mentioned by the Department of Examinations and Credit Accumulation (DECA) of TAR UC are stated as below :

1. Criteria for planning

a. If the number of examiner in a room exceeded the number of invigilator required for the room, the extra examiners are to be shifted to nearby venue.
b. Do not assign duty to all lecturers from the same faculty in the same session in order to leave some lecturers to be totally free for each session to enable them to swap duties among themselves.
c. For FAFB 3 hours or 3 hours 15 minutes papers, assign FAFB lecturers as invigilators up to two third of their load of invigilation duties for the whole examination period.
d. For FEBE double seating (DS) papers, assign to FEBE QS and BD lecturers first.
e. For CNBL papers, assign to CNBL lecturers first.

f. Do not assign duty to Muslim Male staff on Friday PM session.
g. Assign duty up to maximum load of duty Z.
h. The number of invigilators in the venue must not exceed the required number.
i. Invigilators must not have more than one duty on same day and same session.
j. Invigilators must not have more than one evening session duty.
k. Invigilators must not have more than one Saturday session duty.
l. Assign up to two third of the load of duty to the lecturers who are doing PHD.

2. Relief and Quarantine Invigilators

a. First name that appears in the relief list of the session must be an experienced invigilator.
b. 60% of the total relief invigilator per session must be experienced lecturer.

3. Chief Invigilators

a. For FAFB 3 hours or 3 hours 15 minutes papers, assign to FAFB chief invigilators up to two third of their load of chief invigilation duties.
b. For FEBE double seating (DS) papers, assign to FEBE chief invigilators first.
c. For block H, if more than 9 rooms are used, the block will be divided into 2 sub-blocks.

3.3 Tabu Search for Invigilation Duty Scheduling

3.3.1 Description

By using Tabu Search algorithm, the invigilation duty scheduling problem will be solved through the swapping within the duties that are violating the hard constraints. The process will begin with an initial solution, and followed by the process of searching and swapping until all of the invigilation duties satisfy the hard constraints.

Given a set of invigilation duties, each of the duty will represent a slot for the assignation of staff to invigilate a specific examination venue during a specific session. Each duty will be checked against the relevant constraints which are imposed to it. At the end, each duty will be assigned with one staff who are able to satisfy all of the relevant constraints.

The flowchart of the overall process of the invigilation duty scheduling is shown as below :

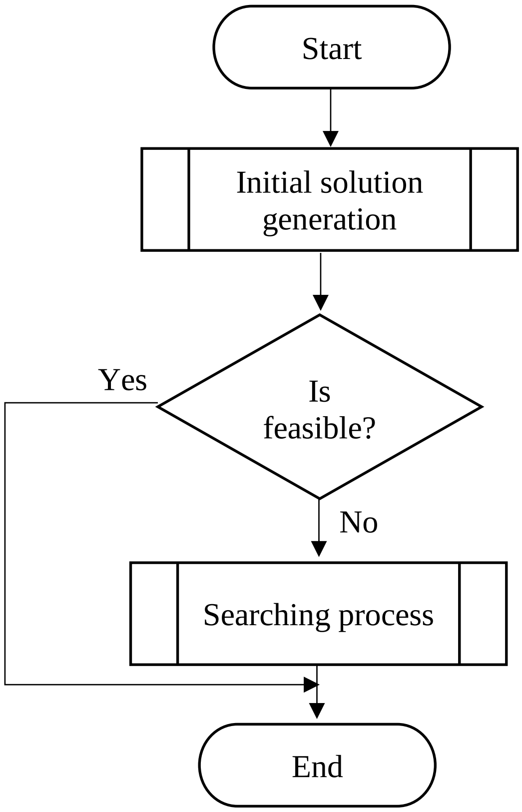

Figure 3.2 Overall process of the invigilation duty scheduling

3.3.2 Elements and Features of Tabu Search

1. **Tabu List**

   Tabu List is a mechanism that used to store the recent move that have to be treated as a forbidden move in the next *n* iterations. This mechanism is useful in avoiding from cycling. In this study, a Tabu List with a size of 20 will be used during the searching process. For every move, the staff who are being swapped will be stored into the Tabu List along with the invigilation duty which was previously assigned to the particular staff. The item that was being stored in the Tabu List will be called as tabu. When the Tabu List is full, the most previous move will be remove from the list while the most recent move will be added into the list. During the searching process, these forbidden moves will not be considered for the next move until their tabu status are cleared.

2. **Violation List**

   Violation List is a mechanism that used to store all the invigilation duties which have violated any of the constraints. In this study, there are two types of violation list that will be introduced which are the hard violation list as well as hard and soft violation list. The hard violation list is used to stored the invigilation duties which only violated the hard constraints, while the hard and soft violation list is used to store the duties which violated both hard and soft constraints. During the searching process, the most violated process will be searched from the hard violation list first before the hard and soft violation list is being searched.

3. **Cost Function**

   The cost function evaluate the quality of the solution by indicating the extent of violation on the constraints that have been stated. During the scheduling process, the violation score will be calculated for each invigilation duty and the total of the violation scores will be the cost function. The violation score of every hard constraint violation will be higher than the sum of all the scores of the soft constraints violation. The violation score for the violation of every constraints can be different according to the important level of the constraint. In this study, the violation score for the hard constraints range from 10000 to 50000.

4. **Aspiration**

   The aspiration value is a value which could allow the tabu status of a tabu to be ignored. If a move that existed in the Tabu List is able to result in a cost function value that is better than the current best cost function in overall, the tabu status will be overridden and the move will be available.

5. **Neighbourhood Relation**

   The neighbourhood of a solution is the solution that are generated by carrying out one move from the previous solution. In every iteration of the searching process, a set of neighbourhood will be chosen generated for the purpose of determining the best move for the particular iteration. In this study, the move will be the swapping of staff between two invigilation duties, which means that the staff of the chosen duty will be moved to another duty while the neighbour staff will be moved to the chosen duty. Therefore, the set of neighbourhood will be generated by moving the staff of the most violated duty to all of the invigilation duties for evaluation purpose in order to determine which of the duty suit the particular staff the most.

3.3.3 Initial Solution

An initial solution does not necessary to be feasible, but the quality of an initial solution is an important aspect to be concerned with. This is because an initial solution with good quality will greatly reduce the computational time for the invigilation duty scheduling process. However, the time spent on the generation of an initial solution should not be excessively long as it will affect the overall computational time as well. Therefore, neither the quality of the initial solution nor the computational time for an initial solution should be overlooked. Instead, the initial solution should be generated with an appropriate quality within the lowest possible of time.

In this study, the initial solution of the invigilation duty scheduling problem will be generated with an acceptably good quality without consuming a large amount of time. In order to minimized the computational time, the staff are being classified into groups according to some of the important constraints. As the invigilation duty consists of chief invigilator duty and normal invigilator duty, the staff will be separated into two groups according to their category of invigilation at the beginning. After that, the invigilation duty will be assigned with staff one after another according to the category of invigilation duty. The initial allocation of staff will ensure the feasibility of the solution, which mean that all of the constraints will be checked before the staff is being assigned to the particular duty. On the other hand, the invigilation duties that failed to be assigned with feasible staff will be assigned with staff randomly in further allocation. During the allocation process, the staff who reach the maximum load of duty will be eliminate from the choice of allocation to ensure that the number of duty that are assigned to every staff are balanced. The flowchart of the initial scheduling process is shown as below :

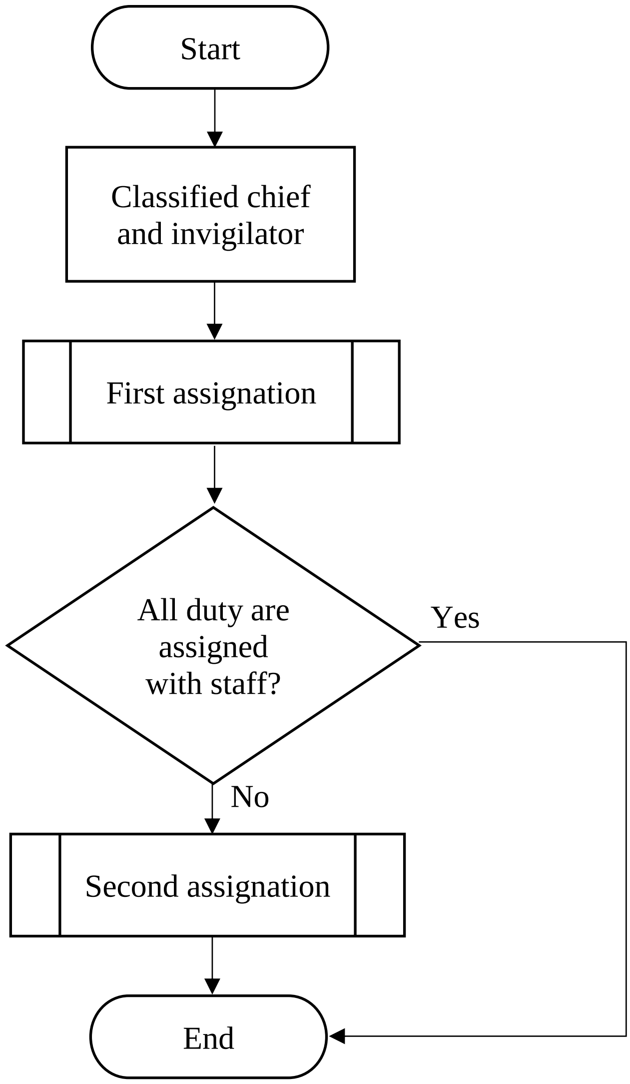

Figure 3.3 General view of initial solution generation

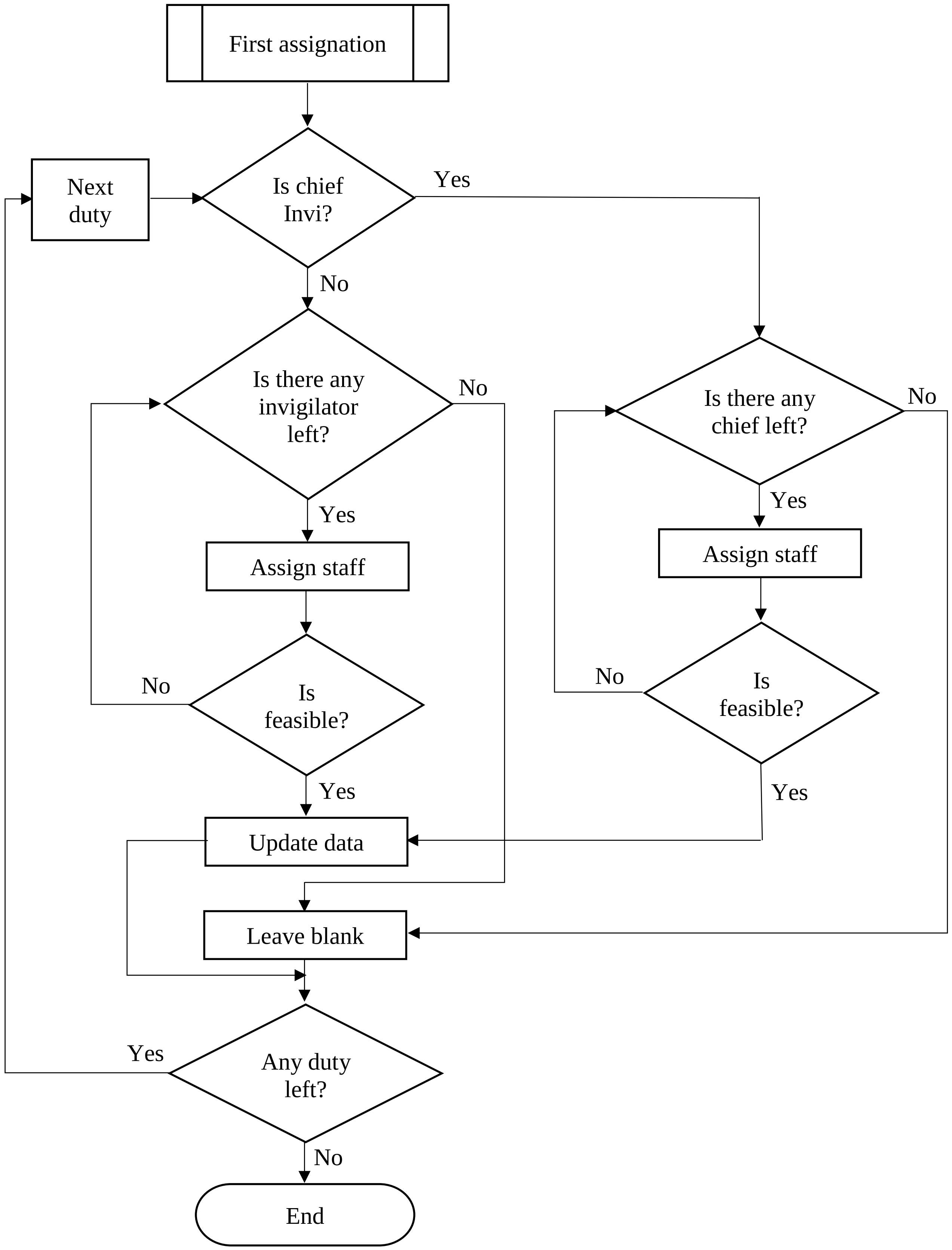

Figure 3.4 First assignation process

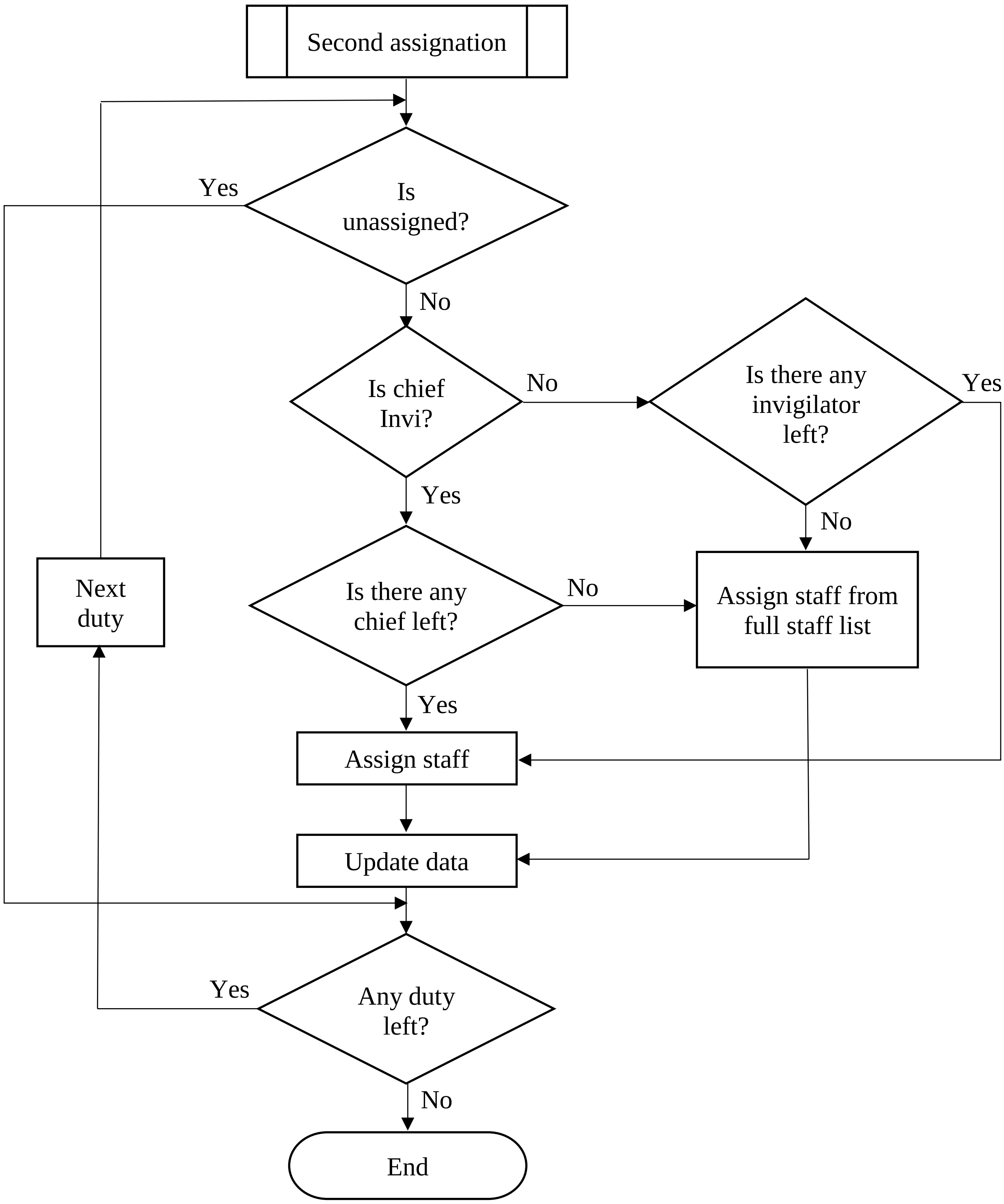

Figure 3.5 Second assignation process

3.3.4 Search Process

Before entering the searching process, the violation score for each of the invigilation duty will be updated in order to calculate the cost function for the purpose of determining the feasibility of the solution. If the initial solution is infeasible, the searching process will be started after the violation lists have been generated.

In the beginning of the process, the most violated invigilation duty will be chosen from the violation lists, which mean that the duty with the highest violation score will be selected. Next, the most critical procedure, which is the searching of best neighbourhood will be carried out after determining the invigilation duty to be swapped. In this process, the staff of the most violated duty will be moved to all of the invigilation duty in order to generated a set of neighbourhood. Each element of the neighbourhood will be evaluated for the purpose of determining the best neighbourhood of the particular iteration. Before the best move is chosen, the tabu status of the best move will be checked so that the forbidden move will not be chosen. However, if the best move is a tabu, the overall cost function of making the move will be evaluated. The tabu status of the particular move will be cleared only if it gives a result which is better than the current best, else, the second best move will be considered.

After the best neighbourhood has been chosen, the swapping process will be conducted and followed by the updating of the data which are related to the staff, invigilation duty as well as the faculty. In order to minimize the computational time, only the violation score of the affected duties will be updated instead of updating the violation score of all of the duties. The cost function of the solution will be evaluate to determine the feasibility of the solution. The searching process will be continued until a feasible solution is generated, else, the tabu list and violation lists will be updated for further iterations. The flowchart of the searching process is shown as below :

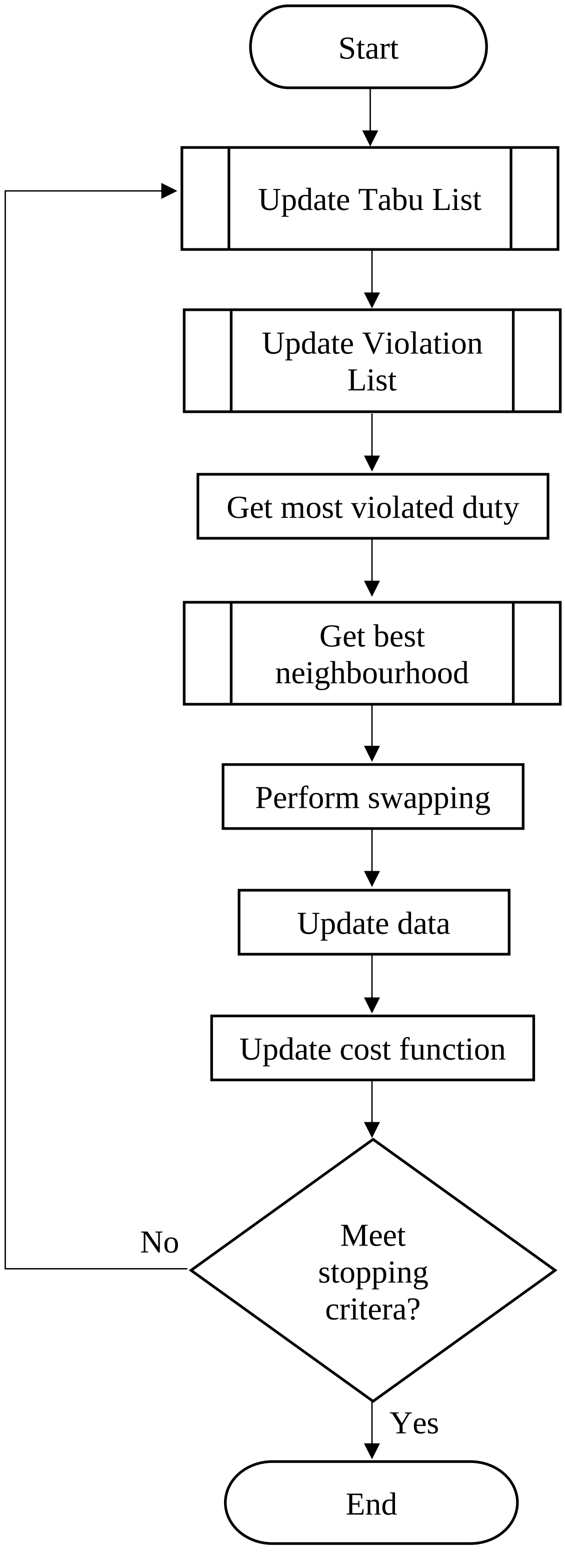

Figure 3.6 General view of searching process

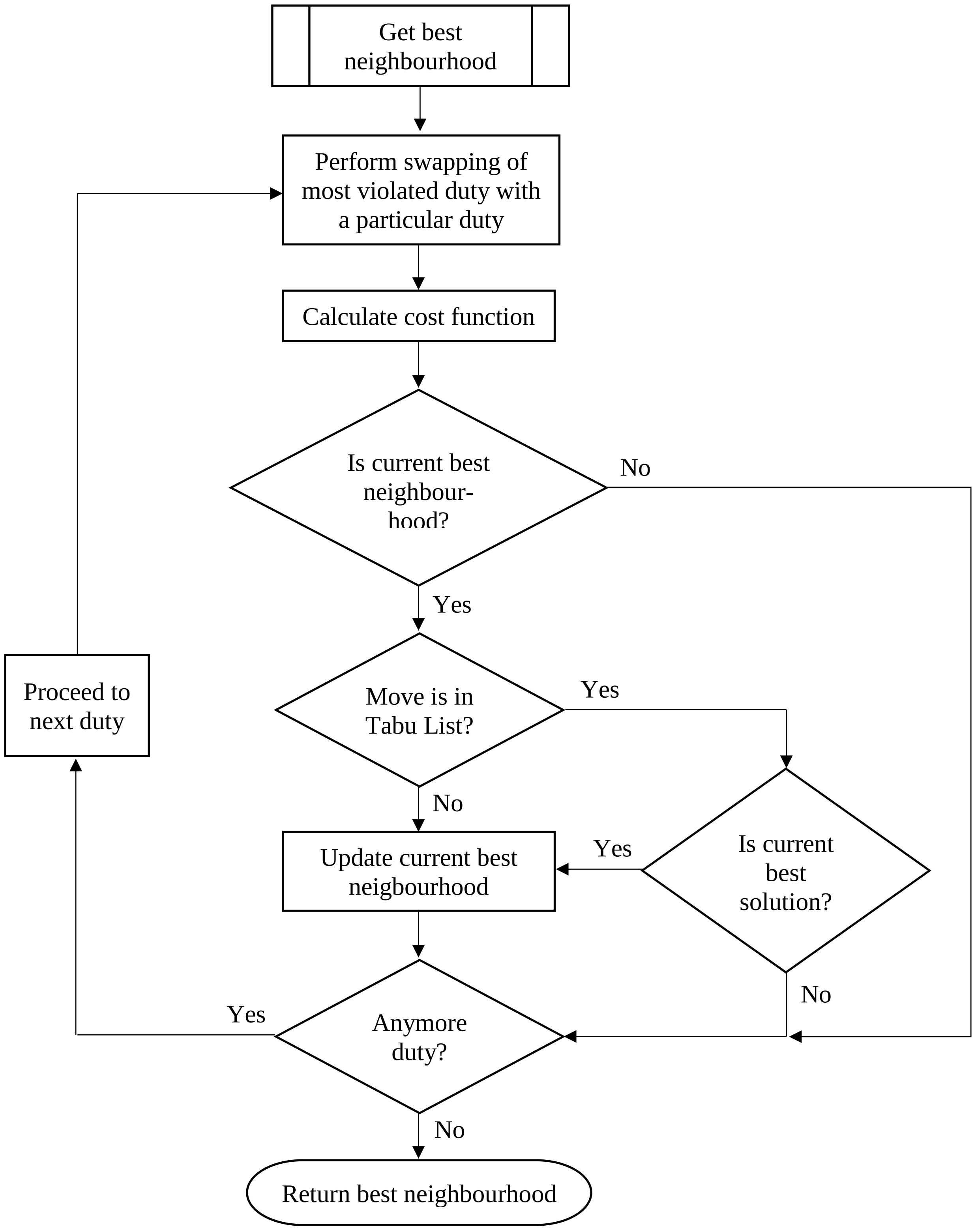

Figure 3.7 Process of getting best neighbourhood

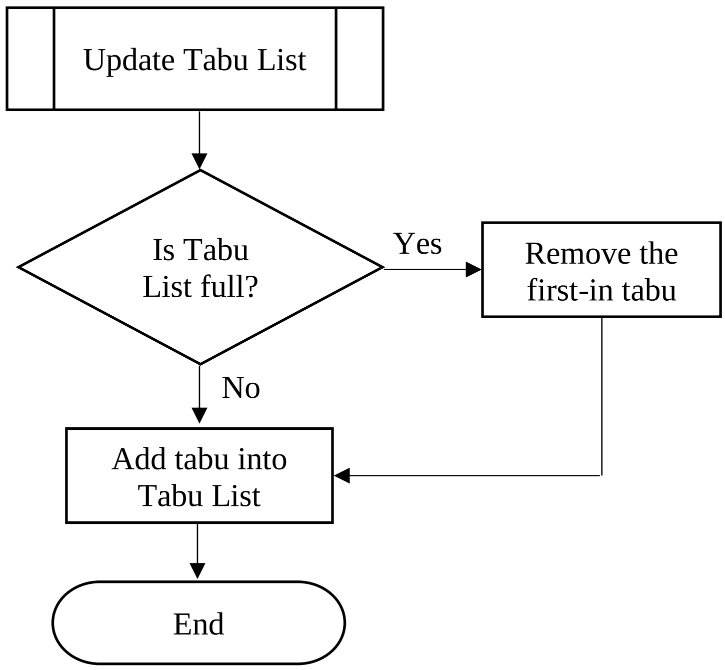

Figure 3.8 Process of updating Tabu List

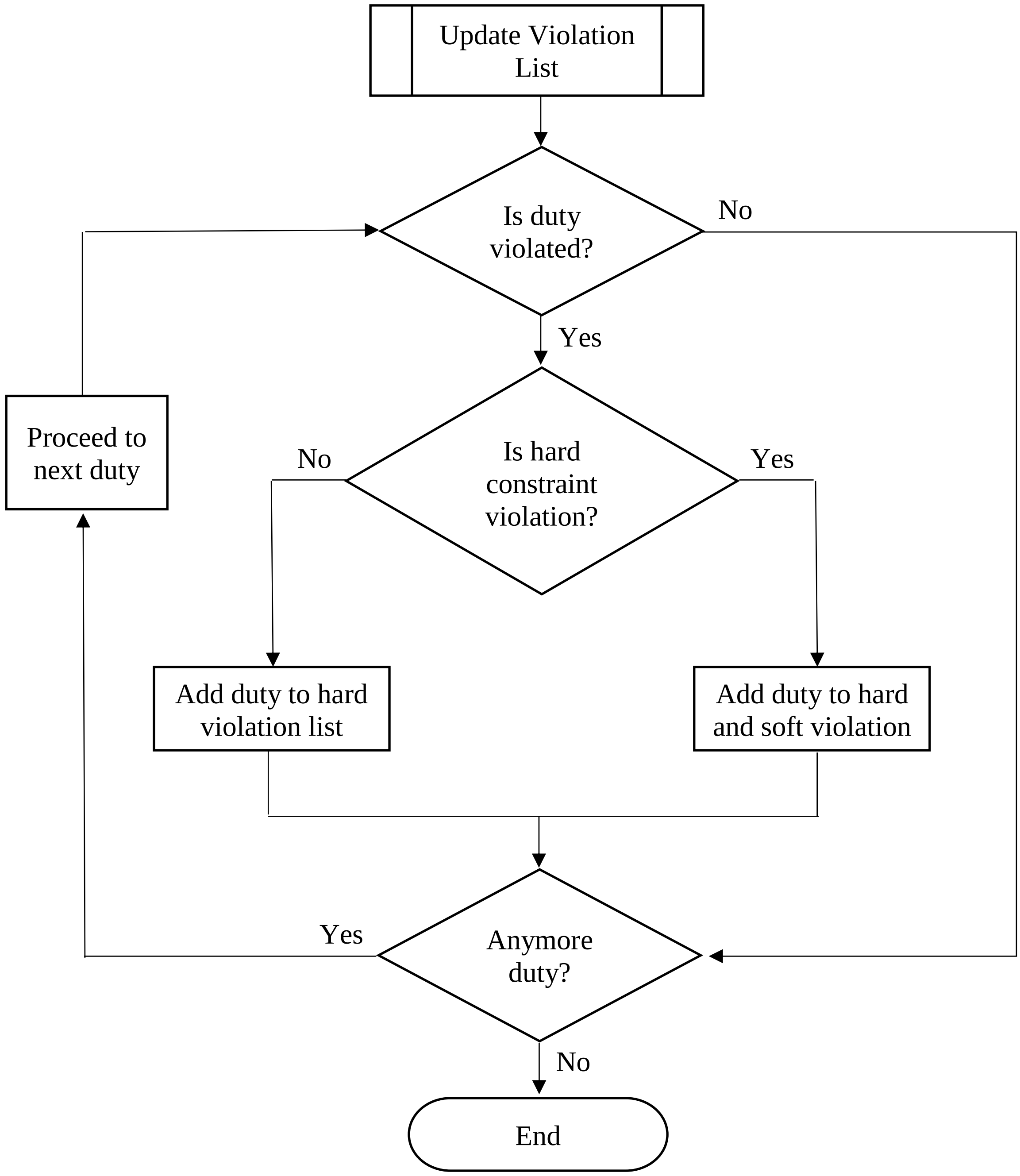

Figure 3.9 Process of updating violation list

3.3.5 Stopping Criteria

In this study, the stopping criteria of the searching process will be as followings :

1. **The solution is feasible, which mean that all of the hard constraint are satisfied.**

   The cost function will be calculate at the end of every iteration to evaluate the feasibility of the solution. Since the lowest violation score for the violation of a particular hard constraint is 10000, therefore, the searching process will be terminated when the cost function is less than 10000.

2. **The number of iterations reached 1000.**

   The number of iterations will be counted along the process of searching and swapping. The process will be terminated even though the solution is infeasible if the number of iterations reached 1000.

3.4 Tools and Materials

3.4.1 Hardware

1. **Laptop**

   Laptop is used for all of the phases which include the stage of system design, system implementation as well as system testing.

3.4.2 Software

1. **Visual Studio 2017**

   The Visual Studio 2017 will be used for the implementation of Tabu Search on the invigilation duty scheduling problem. It is a useful to generate and debug the code of the examination timetabling system. The software will be also used for the system testing after the system is created.

2. **SPSS**

   SPSS is used to generate statistical data for the purpose of conducting relevant test of hypothesis. For example, the relationship between the size of Tabu List and the average computational time will be tested with the simple regression model by using SPSS.

3. **Microsoft Word**

   Microsoft Word is used for the phase of system design which included designing the algorithm of the invigilation duty scheduling as well as the design of flowchart of the scheduling part in the invigilation module.

3.4.3 Mathematical Tools

1. **Line Graph and Bar Graphs**

   Line graphs will be used to show the consistency of the performance of a particular algorithm. On the other hand, bar graph will be used to display the efficiency of the algorithm as well as comparing the performance of different algorithms.

2. **Tables**

   Tables of information will be generated in order to display all of the relevant results as well as statistical information.

3.5 Method of Analysis

3.5.1 Analysing the Efficiency of Tabu Search in Invigilation Duty Scheduling

Tabu Search will be applied on the invigilation module of the examination timetabling system of TAR UC. The system will be tested for five times and the computational time will be recorded. Besides that, some relevant stastistical information such as the mean and the standard deviation of the computational time will be calculated for evaluation purpose. The efficiency and consistency of the algorithm will be evaluated and analysed by using the information gathered.

3.5.2 Analysing the Effect of the Size of Tabu List on Computational Time

Tabu Search with various size of Tabu List will be applied on the invigilation module of the examination timetabling system of TAR UC. The system will be tested for three times for each of the size of Tabu List and the computational time will be recorded. Besides that, some relevant stastistical information such as the mean and the standard deviation of the computational time will be calculated for evaluation purpose. The effect of the size of Tabu List on the computational time will be analysed by using the information gathered.

3.5.3 Analysing The Performance of Tabu Search and Current Algorithm

The Tabu Search based system and the previous system which is generated with other algorithm will be tested in the same condition. Both of the system will be tested for five times and the computational time will be recorded. The relevant statistical information such as the mean and the standard deviation of the computational time will be calculated for both of the system for comparison purpose. The performance of both of the system in terms of efficiency and consistency will be evaluated and thus analysed with the gathered information.

CHAPTER 4 Main Results

4.1 Results

4.1.1 Effect of the Size of Tabu List on the Performance of Tabu Search

In this study, the performance of the Tabu Search algorithm with different size of Tabu List which are 20, 40, 60 and 80 was evaluated in order to determine the linear relationship between the size of Tabu List and the performance of Tabu Search. For observation purpose, three samples of computational time were recorded for each of the size of Tabu List and the mean computational time for each of the size of Tabu List was then calculated. The test of each run of the system was conducted by using the same hardware and software under the same condition in order to ensure the validity of the result. The table and bar chart below shows the result of the test which includes the computational time for each run as well as the average computational time for each of the size of Tabu List.

Size of Tabu List	Computational Time (minutes)			Mean Time (minutes)
	Test 1	Test 2	Test 3	
20	136	207	149	164
40	139	146	256	180.33
60	163	137	205	168.33
80	197	132	186	171.67

Table 4.1 Computational Time for Tabu Search with different size of Tabu List

Based on table 4.1, the computational time of the invigilation duty schedule are varying randomly around 190 minutes for each of the test. For the average computational time, the result is fluctuating around 170 minutes. The lowest mean computational time that was achieved in this study is 164 minutes which is came from the implementation of Tabu List's size of 20. On the other hand, the longest average computational time is obtained from the implementation of Tabu List's size of 40. By observing the result of each individual run, the highest computational time can go up until 256 minutes which is obtained from the implementation of Tabu List's size of 40, while the shortest computational time is achieved by the implementation of Tabu List's size of 80 which is only 132 minutes. From the observation on the result of the tests, it can be seen that there is no linear relationship between the size of Tabu List and the performance of Tabu Search as the computational time rises and drops randomly as the size of Tabu List increases.

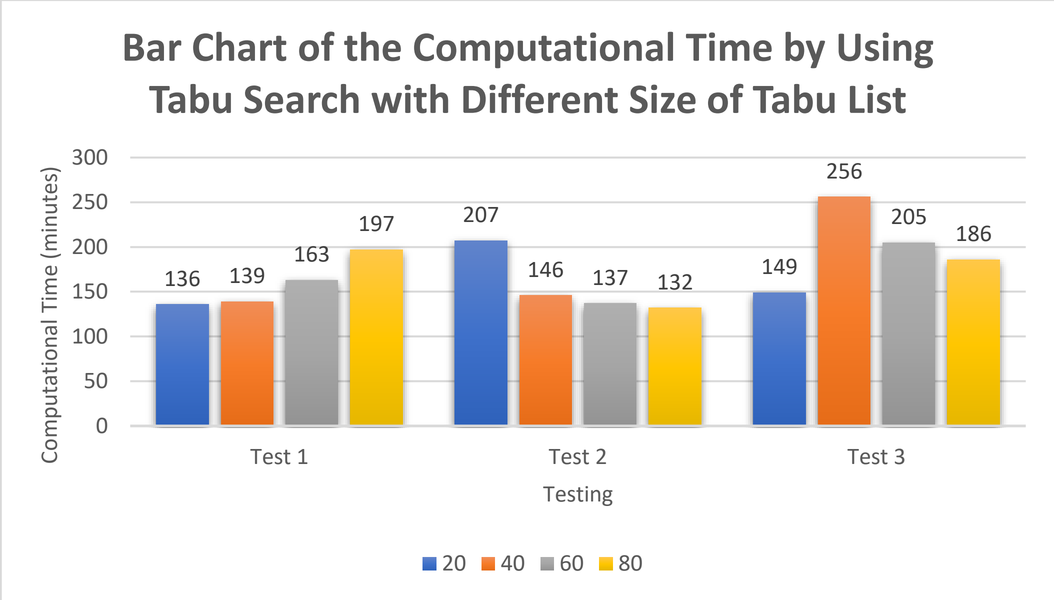

Figure 4.1 Bar Chart of the Computational Time by Using Tabu Search with Different Size of Tabu List

From figure 4.1, we can notice that the results from each of the test are unstable as the performance of Tabu Search with every size of Tabu List fluctuated randomly in three of the tests. This situation shows that the performance of Tabu Search is not linearly related to the size of Tabu List. For the first test, the algorithm with Tabu List's size of 20 performed the best by producing the schedule in 136 minutes, while the longest computational time is obtained from the test of the implementation of Tabu List with size of 80. However, the result of the second test is showing a contrast with the result in the first test, where the longest computational time is obtained from the implementation of Tabu List's size of 20 and the best result which is 132 minutes is achieved from the algorithm with Tabu List of size 80. In the last test, the implementation of Tabu List's size of 20 is able to give the best performance among the implementation of different size of Tabu List again. Somehow, the algorithm with the Tabu List's size of 40 which have a stable performance in the previous tests shows an unexpectedly bad result in the last test by generating the schedule with an extremely long duration which is 256 minutes.

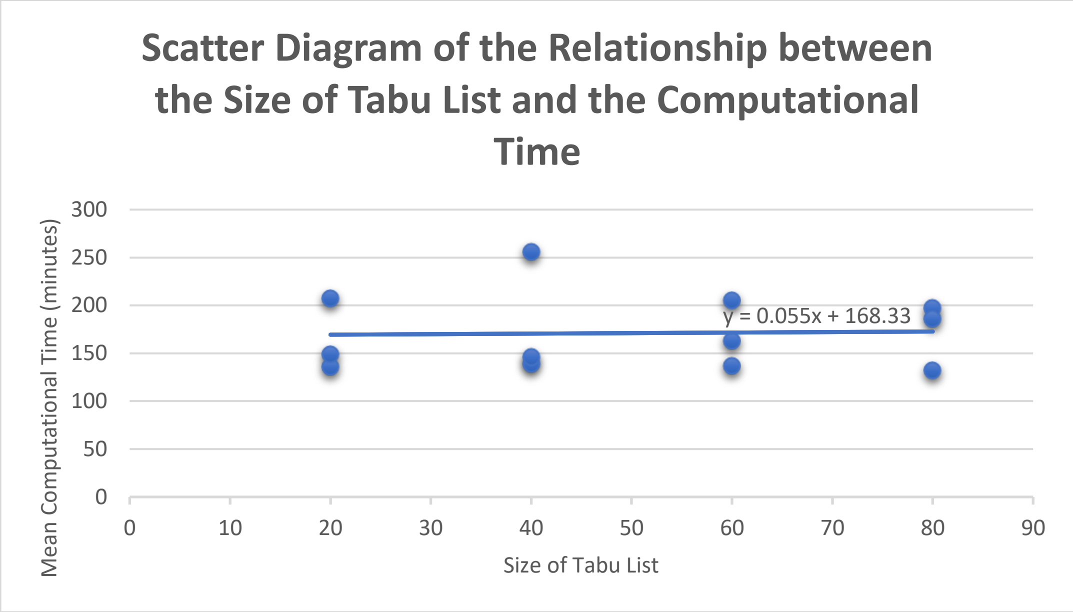

Figure 4.2 Scatter Diagram of the Relationship between the Size of Tabu List and the Computational Time

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.033[a]	.001	-.099	40.815

a. Predictors: (Constant), Size of Tabu List

Figure 4.3 Model Summary

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	18.150	1	18.150	.011	.919[b]
	Residual	16658.767	10	1665.877		
	Total	16676.917	11			

a. Dependent Variable: Computational Time

b. Predictors: (Constant), Size of Tabu List

Figure 4.4 ANOVA Table

Coefficients[a]

Model		Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	168.333	28.861		5.833	.000
	Size of Tabu List	.055	.527	.033	.104	.919

a. Dependent Variable: Computational Time

Figure 4.5 Statistical Information of Coefficients

Besides of observing on the results, the linear regression model is being used to investigate the linear relationship between the size of Tabu List and the performance of Tabu Search in terms of computational time. The relevant statistical information such as the correlation coefficient and the determination of correlation was computed for the purpose of determining the linear relationship between the size of Tabu List and the performance of Tabu Search. Furthermore, a P-test is also being conducted in order to provide a stronger evidence in concluding the relationship between them.

From the scatter diagram in figure 4.2, we can notice that the point are plotted in an unordered manner and most of it are being separated far away from the line of best fit. With this observation, we can state a hypothesis that the linear relationship between the size of Tabu List and the performance of Tabu Search does not exist.

Referring to the model summary in figure 4.3, the coefficient of correlation is 0.033 which proves that the performance of Tabu Search has a very weak linear relationship with the size of Tabu List and it can be considered as nearly absent. Moreover, the coefficient of determination with a value of 0.001 concluded that there is only 0.1% of the variation in the mean computational time is affected by the size of Tabu List. Therefore, Both of this statistical information is able to provide an evidence that the performance of Tabu Search will not be affected by the size of Tabu List linearly.

Lastly, the P-test is being conducted as below:

Hypothesis

H_0 : There is linear relationship between the size of Tabu List and the performance of Tabu Search.

H_1 : There is no linear relationship between the size of Tabu List and the performance of Tabu Search.

Critical Region

Given a = 0.05, Critical Value = a = 0.05

Critical Region : p-value > 0.05

P-Value

From figure 4.5, p-value = 0.919

Conclusion

Since the p-value is 0.919 > 0.05, reject H_0 at significant level of 0.05 and it can be conclude that there is no linear relationship between the size of Tabu List and the performance of Tabu Search.

Based on the results as shown above, we can conclude that the computational time is not linearly related to the size of Tabu List. Therefore, the performance of Tabu Search will not be affected by the size of Tabu List in a linear way.

4.1.2 Performance of Tabu Search

System	Computational Time (minutes)					Mean	Standard Deviation
	Test 1	Test 2	Test 3	Test 4	Test 5		
Tabu Search Based System	136	207	149	172	215	175.8	34.74

Table 4.2 Computational Time for Tabu Search with Tabu List' size of 20

In this study, the implementation of Tabu List's size of 20 is being used for the analysis of the performance of Tabu Search in terms of computational time. Thus, there are two additional tests that have been carried out in order to collect five samples of computational time for the implementation of Tabu List's size of 20. The average computational time is then recalculated again.

According to the results in this study, Tabu Search is able to generate an invigilation duty schedule within 175.8 minutes on average. The fastest record of Tabu Search in generating the invigilation duty schedule is achieved in the first test which is 136 minutes. This result is considered fast enough for the invigilation duty scheduling. On the other hand, the worst performance of it in this study is given in the last test as the result rises to 215 minutes. However, this result is still reasonable as the problem of invigilation duty scheduling is complicated and it might require a large amount of time.

By referring to the range of the result, the range of 79 minutes indicates the inconsistency of the algorithm in generating a schedule within a specific duration consistently. However, the standard deviation of the result is 34.74 minutes which mean that the results of each test will only have a difference of 34.74 minutes with the mean duration on average. Therefore, the consistency of the algorithm in terms of computational time can be considered acceptable as the computational time for each schedule does not vary in a large extent.

In this investigation, the performance of Tabu Search is acceptably good in terms of efficiency in generating the invigilation duty schedule. However, the performance of the algorithm is slightly unstable as it might take longer time in generating a schedule in some of the time.

4.1.3 Comparison of Tabu Search Based system and Current System

Lastly, a comparison of the performance has been made between Tabu Search algorithm and the current algorithm that is being used for the current system. For comparison purpose, five test run of the current system has been conducted in order to collect the sample data of the computational time for the current algorithm. The relevant statistical information has been calculated for the aim to compare the efficiency as well as the consistency of the algorithms in solving the invigilation duty scheduling problem. The results is tabulated as below :

System	Computational Time (minutes)				
	Test 1	Test 2	Test 3	Test 4	Test 5
Tabu Search Based System	136	207	149	172	215
Current System	933	867	866	834	874

Table 4.3 Computational Time for Tabu Search and the Current Algorithm

System	Mean Time (minutes)	Standard Deviation	Longest Time (minutes)	Shortest Time (minutes)
Tabu Search Based System	175.8	34.74	215	136
Current System	874.8	36.02	933	834

Table 4.4 Statistical Information of the Performance of Tabu Search and the Current Algorithm

For the efficiency of the algorithm, we are able to notice the huge difference between Tabu Search and the current algorithm in all of the tests, where the computational time of the current algorithm in each test is much larger than Tabu Search algorithm. By referring to the mean computational time for both of the algorithms, Tabu Search which has a mean duration of 175.8 minutes shows a better efficiency in solving the invigilation duty scheduling problem as compared to the current algorithm which generates the schedule in 874.8 minutes on average. The best result that has been achieved by Tabu Search is 136 minutes which is 698 minutes faster than the shortest computational time that can be achieved by the current algorithm. On the other hand, the longest computational time for Tabu Search among these test is only 215 minutes, while the computational time for the current system can rise up to 933 minutes which is an unfavourable result.

For the consistency of the algorithms in terms of computational time, the standard deviation of the results is being used for the comparison for the algorithms. There is only a slight difference between the standard deviation of the results of Tabu Search and the current algorithm, where the standard deviation for the result of Tabu Search is 34.74 minutes while the result of the current algorithm has a standard deviation of 36.02. Therefore, the information shows that the consistency of Tabu Search in solving the invigilation duty scheduling problem is slightly better than the current algorithm.

Based on the result, it can be concluded that Tabu Search performs better than the current algorithm in the invigilation duty scheduling as Tabu Search algorithm is able to generate the invigilation duty schedule in a more efficient and consistent way as compared to the current algorithm.

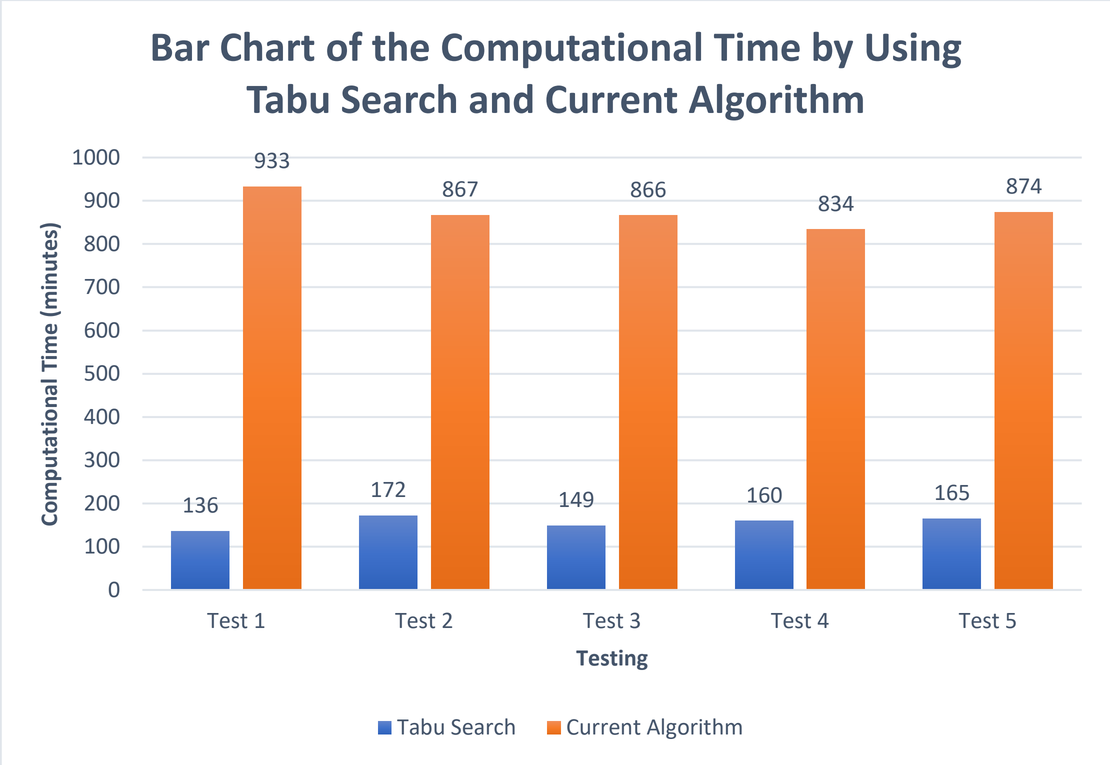

Figure 4.6 Bar Chart of the Computational Time by Using Tabu Search and Current Algorithm

The bar chart as shown in figure 4.6 is used to show the comparison of the efficiency of the algorithms in invigilation duty scheduling visually. From the chart, it can be clearly seen that Tabu Search is able to outperform the current algorithm in solving the invigilation duty scheduling problem with a great extent as there is a big gap in their computational time in all of the tests.

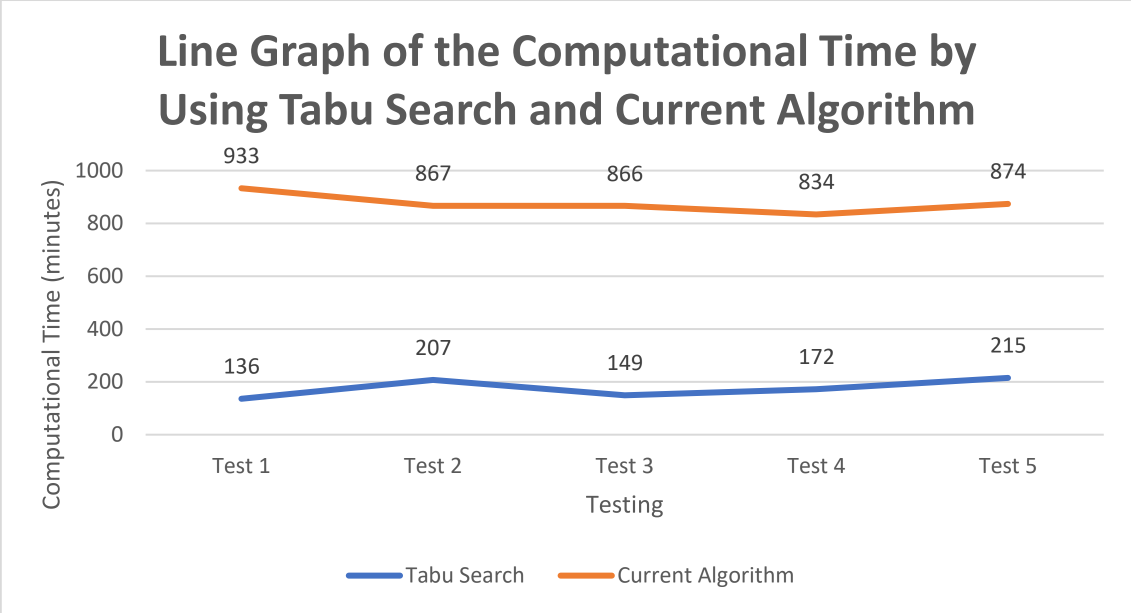

Figure 4.7 Line Graph of the Computational Time by Using Tabu Search and Current Algorithm

The line graph as shown in figure 4.7 is used to show the comparison of the consistency of the algorithms in generating the invigilation duty schedule visually. From the line graph, we can notice that the performance of Tabu Search in the first three test is slightly unstable. However, it can be concluded that both of the algorithms has similar consistency in solving the invigilation duty scheduling problem in overall.

4.2 Discussion

4.2.1 Effect of the Size of Tabu List on the Performance of Tabu Search

Based on the investigation in this study, we can conclude that the size of Tabu List does not affect the performance of Tabu Search in solving the invigilation duty scheduling problem in a linear way. However, the effect of Tabu List's size on the performance of Tabu Search could be affected by the constraints of the optimization problem as well as the set of data. In this study, the constraints that have been stated does not strictly restrict the assignation of the resources to the duties, which mean that every duty in the data set could have a large number of candidate staff that can be assigned. This situation might greatly reduce the possibility of cycling as the recent moves will not be made continuously. As a result, the size of Tabu List will not affect the performance of Tabu Search in generating the schedule as it becomes larger. Therefore, the implementation of large size of Tabu List will not give any advantage to the performance of Tabu List in this problem and thus, Tabu List with size of 20 is considered as the most appropriate choice in this study.

However, the investigation has only been made on the linear relationship between the size of Tabu List and the performance of Tabu Search in this study. Somehow, there might also be any other relationship between them, which mean that the size of Tabu List could affect the performance of Tabu Search in other way in solving the problem of invigilation duty scheduling. Thus, a deeper investigation have to be made in order to ensure that the relationship between the size of Tabu List and the performance of Tabu Search does not exist.

4.2.2 Performance of Tabu Search

In this study, the standard version of Tabu Search is implemented to the problem of invigilation duty scheduling for the purpose of analysing the performance of Tabu Search in solving this particular optimization problem. From the observation on the computational time, we can conclude that Tabu Search is able to solve the invigilation duty scheduling problem efficiently as the Tabu Search based algorithm is able to generate an invigilation duty schedule within a short duration which is around 2 to 3 hours. This duration is considered sufficiently fast as we take the complication of the problem into consideration.

Despite of the high efficiency in solving the problem, the consistency of the in performing well in the problem is not able to meet the expectation as the computational time required to generate a schedule is slightly unstable. However, the poor consistency might be caused by the inconsistency of the quality of the initial solution which is due to the algorithm of generating the initial solution instead of the Tabu Search algorithm. The poorer the quality of the initial solution, the larger the number of iteration have to be carried out during the searching process which will be implemented with Tabu Search. Consequently, the computational time to complete the whole scheduling process will be longer when an initial solution with lower quality is generated.

However, the computational time of the algorithm will be differed as the number of constraint and the resources change. The larger the number of constraints to be involved, the longer the computational time that required to generate a schedule. On the other hand, the computational time will also decrease as the number of resource tends to be sufficient. Therefore, the performance of Tabu Search can only be analysed and evaluated based on the results that have been generated with certain data set as there is no any standard criteria that can be used to evaluate the performance of an algorithm. In order to obtain a better evaluation on the algorithm, comparison of the performance among different algorithms has to be made for the aim to determine the suitability and performance of the algorithms in a specific optimization problem.

4.2.3 Comparison of Tabu Search Based system and Current System

In the end of the study, the performance of Tabu Search and the current algorithm has been compared with each other in order to determine the algorithm that is more suitable to be implemented in the invigilation duty problem. The result of the comparison shows that the performance of Tabu Search in solving the invigilation duty scheduling problem is much better in terms of the efficiency as well as consistency as compared to the current algorithm.

From this study, the huge difference between the computational times for both of the algorithms could show that the efficiency of Tabu Search in solving the invigilation duty scheduling problem is outstanding as compared to the current algorithm. By applying Tabu Search to the problem, the TAR UC will be able to save up around half a day to generate an invigilation duty schedule. This could be a very significant change in the efficiency of the system and it would bring a lot of benefit to the particular institution. One of the main advantages that could be obtain is the great reduction in the time and cost that will be required in generating the schedule. Besides that, the short computational duration will be able to bear with the risk of system break down as the schedule can be generate within a short period right after the system is recovered.

Apart from that, Tabu Search also has a better consistency in terms of computational time as well as the quality of the solution as compared to the current algorithm. Based on the result, Tabu Search is slightly better in maintain the efficiency of the algorithm in solving the invigilation duty scheduling problem. Furthermore, gap between the consistency of Tabu Search and the current algorithm could be enlarged if the algorithm of generating the initial solution for Tabu Search could be enhanced. Besides of the consistency of the algorithm's efficiency, the current algorithm also failed to maintain the quality of the result in an acceptable level for every scheduling that have been conducted. This situation could not be overlooked as the feasibility of the schedule is one of the main concern in this problem. On the other hand, Tabu Search is able to ensure the feasibility of the schedule if the initial solution could be generated well.

However, despite of the high efficiency of Tabu Search in generating the invigilation duty schedule, the quality of solution generated with Tabu Search algorithm will be slightly inferior to the current algorithm. In contrast to Tabu Search, the current algorithm will allocate staff to each invigilation duty one after another after finding the best staff to be assigned to the specific duty. Although the quality of the solution in the beginning phase will be outstanding, but the quality of the solution will drop as the process continues to run. Eventually, the quality of the result will only be slightly better than the solution generated by Tabu Search despite of the large consumption of time in generating the schedule. Furthermore, a sufficiently good solution, which is able to satisfy all of the hard constraints and some of the soft constraints of the problem, should be acceptable for the invigilation duty scheduling problem. Therefore, Tabu Search will still be the better choice as compared to the current algorithm after taking all of these aspect into consideration.

CHAPTER 5 Conclusion

5.1 Conclusion

In a nutshell, it can be concluded that Tabu Search is a suitable algorithm to be applied in the invigilation duty scheduling as it can consistently provide a good performance in generating the invigilation duty schedule in terms of computational time. Besides that, it does not overlook the importance of the quality of the solution by providing a feasible schedule that fulfil all of the hard constraints that have been stated. This study has shown the high flexibility of Tabu Search algorithm as it is able to be suited in various type of optimization problems including the invigilation duty scheduling problem. With the potential of Tabu Search in solving the invigilation duty scheduling problem, this topic should be further investigated in order to figure out a better strategy in applying the Tabu Search by modifying the features and elements of it during the application. This is because the appropriate use of Tabu Search could lead the performance of the system to a higher level. Therefore, the application of Tabu Search in the invigilation duty scheduling can be considered after making appropriate modification to the algorithm so that the examination timetabling system will be able to generate a high quality invigilation duty schedule in an efficient way.

As compared to the current algorithm, Tabu Search algorithm is more suitable to be used in solving the invigilation duty problem according to the result of the study. This study has shown the high efficiency and performance of Tabu Search in solving the particular problem which proved that it has better potential for this problem, and therefore, it should be a better choice than the current algorithm. From the result, Tabu Search seems to be able to provide a feasible solution within a shorter period and the quality of the solution generated with Tabu Search is not inferior to the solution provided by the current system. Besides that, it is not necessary to spend an excessively long period to generate a perfect solution for the problem as an educational institution normally required only a feasible invigilation duty schedule with acceptable quality instead of a perfect timetable. Therefore, Tabu Search algorithm would be a more appropriate choice for the invigilation duty scheduling problem.

Last but not least, this study has also provided evidence that the performance of the algorithm does not have a direct relationship with the size of Tabu List. Therefore, the use of Tabu List with smaller size which is the size of 20 will be appropriate as it will not gives any positive effect to the performance of the system even if a larger size of Tabu List is implemented. This could help to prevent the operating system from allocating excessive and unnecessary memory space to the program.

5.2 Achievement and Contribution

The application of Tabu Search algorithm in solving the invigilation duty scheduling problem has provided small but valuable contribution to the relevant field of this study. In overall, this study has successfully achieved the main goal as well as some of the secondary goals which have been determined in the early stage of the project.

First of all, this study is able to give more information related to the efficiency and consistency of Tabu Search in solving the invigilation duty scheduling problem which could provide guidance to the developers of the relevant system in choosing an appropriate algorithm. By referring to the result of the study, they will be able to determine the suitability of the particular algorithm in solving the problem. Furthermore, it can also be used for the purpose of comparison if there is any application of other algorithms that have been conducted in solving the similar problem. Consequently, this contribution will indirectly help the educational institution in minimizing the cost in terms of time for the generation of invigilation duty as the right choice of algorithm will greatly reduce the computational time of a schedule. Therefore, the invigilation module in the examination timetabling system in TAR UC could be improved in a large extent as the Tabu Search is proved to be more suitable than the current algorithm in this study.

Apart from that, this study has also determined the relationship between the size of Tabu List and the average computational time of the invigilation duty schedule. Through this study, we could get to know about the effect of the size of Tabu List to the performance of invigilation module. Thus, we will be able to determine an appropriate size of Tabu List to be implemented in order to optimize the performance of the system in terms of computational time as well as resource utilisation.

Last but not least, this study can also act as a reference for the developers who are trying to implement Tabu Search algorithm in solving various kinds of optimization problems. From this study, they will be able to acquire a better understanding on the basic principle of Tabu Search as well as the way of implementing this algorithm to a particular problem. Besides that, this study could also indirectly assist the developers in generating a suitable Tabu Search based algorithm for a particular problem by providing some information regarding to the features and elements of Tabu Search and thus, helping them in modifying the elements of the algorithm. Therefore, this could be a useful material which could provide hints for them in the application of Tabu Search in other optimization problems especially during the phase of system design.

5.3 Limitations and Future Studies

5.3.1 Limitations

Although the objectives of the project have been achieved, there is also some imperfections that were caused by the issues as mentioned above. In this study, the quality of the solution failed to meet the expectation as the aspect of soft constraint satisfaction has been slightly overlooked in order to minimize the computational time. In fact, the performance of the invigilation module should be enhanced without overlooking the quality of the solution, but the invigilation module which was developed in this study failed to fully fulfil the soft constraints that have been stated. However, the quality of the solution is considered acceptable as it is able to satisfy all of the hard constraints as well as some of the soft constraints.

Besides that, due to the lack of time, a deep research on the Tabu Search algorithm could not be done in a short period of time which causes the implementation of the algorithm to be imperfect. The time limitation does not allow the investigation on all of the features and elements of Tabu Search to be conducted. In this study, the time length for the project is only sufficient for the process of investigation on the effect of size of Tabu List to the performance of the algorithm. In consequence, this study is not able to design the most suitable Tabu Search based algorithm for the invigilation duty scheduling problem.

Apart from that, the possibility of the solution to be feasible is highly dependent on the generation of the initial solution. In this case, longer computational time have to be spent on the phase of generating the initial solution in order to ensure that the combination of staff that assigned to the duties is able to form a feasible schedule at the end of the scheduling process. As a result, the overall time that required to generate a feasible invigilation duty schedule will be increased. However, this study failed to find out the best way to generate the initial solution. Therefore, the system generated in this study is only able to perform moderately in the initial phase in terms of the computation speed and the consistency of the solution's quality.

5.3.2 Future Studies

In the future studies, there will be a lot of work that have to be done in order to optimize the performance of Tabu Search in solving invigilation duty scheduling problem. First of all, the quality of the solution will be improved in such a way that the computational time is able to maintain in an acceptable level. Besides of satisfying the hard constraints, the soft should also be taken into consideration as well for the aim of improving the user's satisfaction. In order to achieved this goal, a thorough investigation on the Tabu Search based algorithm have to be conducted so that appropriate modification can be made.

Besides that, investigations on the features and elements will be carried out in order to achieved a better understanding on the implementation of Tabu Search on the invigilation duty scheduling problem. The elements such as Tabu List, aspiration, searching strategy as well as the calculation of cost function will be analysed and investigated thoroughly for further modification to be made. Through these investigations, we will be able to understand the effect of various strategy of implementing Tabu Search and thus finding out the best and most suitable way to implement Tabu Search on a particular problem. As a result, the performance of the algorithm in solving the particular problem will be enhanced to a greater level without affecting the quality of the solution.

Lastly, the phase of generating the initial solution will also be one of the main concerns in the future study. A better algorithm in generating the initial solution will be figured out in order to achieve a better quality of initial solution within an acceptable period of time. Therefore, research will be done on different kinds of algorithms of generating the initial solution in order to determine the most suitable algorithm in solving the invigilation duty scheduling problem. Furthermore, we will also find out an appropriate strategy which may be more suitable to a specific case and thus making certain changes in the algorithm according to the case of a particular invigilation duty scheduling problem. With this study, the consistency of the algorithm’s performance will be increased besides improving the performance of the algorithm in terms of computation time.

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